1. PROBLEM STATEMENT

More than 10,000 children die every day from hunger [1]. Farmers in the United States are unable to meet the domestic demands for food, with imports exceeding exports by over $41 billion [2]. Additionally, farmland is lacking in nutrients; at least one soil or water resource concern exists on 49% of surveyed soybean, oat, cotton, and wheat fields [3]. The process for testing soil for nutrients is long, requiring multiple individual samples to create a representative composite sample. Depending on the sampling methodology, extension services generally recommend collecting approximately 10–20 cores from each management area or zone [4]. Lab tests cost around $22-$30 [5], [6].

1.1. The Need

Current attempts to address this issue are flawed. The ROGO soil sampling robot follows a pre-programmed grid and collects soil samples to send off to another lab for testing. It is expensive to own and to operate, running on diesel [7]. The Agrobot Lala does in-field analysis, but it can only measure nitrate, takes about 30 minutes per sample, can only run for 3 hours on its battery, is more inaccurate than official lab results, and costs roughly €50,000 [8].

The best way to get accurate and complete measurements of the quality of soil is an official lab test, but testing every point on a plot of land is costly and time consuming, taking up to 15 days to get a result [5]. There is a need for a fast, low-cost way to identify which areas of a field vary enough to justify sending samples to a lab8

1.2. Our Solution

We propose a Digital Instrument for Remote Terrain Testing (D.I.R.T.T.), an autonomous, solar-powered robot that surveys a large plot and identifies areas where soil conditions differ enough to warrant laboratory testing. Instead of replacing the lab, D.I.R.T.T. tells where to send samples from, so that the costly lab analysis is spent where it is most needed. D.I.R.T.T. measures four soil properties, each chosen for a specific agronomic value: pH, electrical conductivity (EC), moisture, and temperature.

1.3. Technical Background

pH governs the availability of nutrients to plants, and it can vary wildly within a single plot of land. One mapping study measured a 2.9-unit range of pH in one field, and uniform lime application (which is the standard way to reduce acidity in soil) would leave over 63% of the land over-limed [9]. Mapping where pH differs shows where acidity problems are concentrated.

Electrical conductivity responds to several soil properties all at once, including clay content, moisture, salinity, cation exchange capacity, and organic content [10]. Because of this, EC maps are widely used to delineate management zones and to guide where soil samples are to be taken, giving a more representative set than traditional grid sampling [10].

Moisture and Temperature are used to gain more accurate EC readings [10], [11]. Moisture is useful for detecting drainage problems, one of the largest problems with soil health [12]. Temperature is useful for detecting biological activity[13].

D.R.I.T.T. is designed to address the cost and operating limitations of existing robots. Where the ROGO robot runs on diesel [7] and the Agrobot Lala costs roughly €50,000 [8], D.I.R.T.T. is solar powered, removing fuel costs and extending the time it can operate in the field.

As D.I.R.T.T. travels the plot, it logs each reading with its GPS location and plots the data of the field. Areas where readings fall outside expected ranges, or change sharply, are highlighted as candidates for lab sampling. The farmer can then collect a small number of samples from those areas, plus a few from areas that look normal to confirm the screening, instead of sampling the whole field.

2. DESIGN REQUIREMENT SPECIFICATIONS

This section outlines the marketing requirements, engineering specifications, constraints, and standards to ensure the success and safety of D.I.R.T.T.

2.1. Features and Requirements

The requirements in this section translate the need identified in Section 1 into measurable targets: a fast and low-cost solution to identify which areas of a field vary enough to justify laboratory soil testing. D.I.R.T.T. screens a field rather than replacing a laboratory. Its position accuracy and measurement tolerance therefore only need to be fine enough to separate one area of a field from another, not to match a laboratory result. Several requirements are also set to exceed the limitations of the existing in-field robots described in Section 1.1, particularly sampling time, operating time, and dependence on fuel. Section 2.1.1 states the features and the system design requirements that quantify them. Section 2.1.2 divides those requirements among the subsystems and describes how each one is demonstrated.

2.1.1. Features and System Design Requirements

Table 2-1 lists five features and six system design requirements, each with a brief justification. The paragraphs following the table explain the requirements that depend on calculations or outside sources.

Table 01: Features and System Design Requirements

Features	System Design Requirements	Justification
1	The robot stops within 5 m of each commanded test point, with test points spaced ≤ 100 m apart across the survey area.	This yields about one point per 2.5 acres, comparable to conventional grid sampling densities [4].
2	Readings agree with reference instruments within ±0.5 pH and ±1 °C.	This tolerance resolves pH differences well within the 2.9-unit range observed in a single field [9].
3	The robot detects loss of the base link within 30 s, ignores lost messages, and returns to the base on its own.	A single dropped message does not abort a survey, but the robot never operates unsupervised for long
1, 3	The robot detects obstacles ≥ 50 cm tall at a distance of ≥ 1 m and stops or steers around them without contact	Fence posts, irrigation risers, equipment, livestock, and people all exceed this height.
4	The robot completes at least 3 hours of survey operation with no solar input	One cloudy day does not end a survey.
5	The base station maps 100% of saved data points onto a map of the area.	This implements the screening method described in Section 1.3, and EC maps are an established basis for delineating sampling zones [10].

Features:
1. D.I.R.T.T. drives itself to each test point in the field.
2. D.I.R.T.T. measures soil pH, EC, moisture, and temperature at each test point.
3. D.I.R.T.T. reports its results to a base station and returns home safely when communication fails.
4. D.I.R.T.T. runs for extended periods on solar power without fuel.
5. D.I.R.T.T. shows the farmer which areas of a field warrant laboratory testing.

The spacing and position-accuracy requirements reflect how survey results are used. A 100 m grid places one test point in each square hectare, or about 2.5 acres, which is comparable to the cell sizes used in conventional grid soil sampling [4]. The difference is that D.I.R.T.T. screens every cell rather than sending every cell's sample to a laboratory. After the survey, a farmer walks to flagged areas with a phone to collect samples.

The measurement requirements follow from the agronomic purpose of each property. pH can vary by 2.9 units within a single field [9], and a ±0.5 tolerance is fine enough to separate acidic areas from the rest of such a field while remaining well short of laboratory precision, which D.I.R.T.T. does not attempt to match. EC responds to clay content, salinity, organic matter, and moisture at the same time [10], and it also changes with temperature by roughly 2% per degree Celsius. A reading taken in a cool morning and one taken in a warm afternoon can therefore differ even when the soil is identical.

The robot may lose connection in a large area of farmland. The 30 s link-loss window tolerates several consecutive lost messages while minimizing unsupervised operation.

Farmlands often have varying obstacles in them, usually sparsely placed. Most of said obstacles are taller than 50 cm; therefore, a 1 m clearance is enough for the robot to navigate around without being obstructed.

The solar requirement is to address the limitation of the Agrobot Lala and ROGO robot. The Agrobot only runs for 3 hours per charge and needs to be manually charged limits its autonomous capabilities [8]. The ROGO robot requiring diesel makes it expensive to run continuously [7]. Operating mostly on solar with a battery backup gives the robot far more autonomy and less upkeep than the other solutions.

The mapping requirement implements the screening method described in Section 1.3: areas whose readings fall outside expected ranges or change sharply between neighboring points become candidates for laboratory sampling. The thresholds are user-set because expected ranges depend on crop, region, and soil type. The farmer then samples the flagged areas, along with a few unflagged areas to confirm the screening, instead of sampling the entire field.

2.1.2. Subsystem Requirements and Demonstration Setup

The robot divides into five subsystems: navigation, control and data, soil testing, drive train and chassis, and power/charging. Table 2-2 assigns each subsystem three testable requirements derived from Table 2-1 and describes the demonstration used to verify each one.

Table 022: Subsystem Requirements and Demonstration Setup

Subsystem/ Item:	Requirements	Demonstration Setup
Navigation	(1) The subsystem must stop the robot within 5 m of the commanded test point. (2) The subsystem must successfully recognize obstacles over 50cm tall ≥ 80% of the time. (3) The subsystem must begin navigating back to the base station within 30s of lost contact.	(1) The robot will be given a coordinate to navigate to. After it has stopped, it will be compared to the actual coordinates it has stopped at (2) A 50 cm tall object will be placed in the path of the robot. It should navigate around it and continue course. (3) The base station will be turned off to simulate the loss of signal, and the robot must automatically return to it.

Control and Data	(1) The subsystem must maintain the link to the base station at ≥ 500 m line of sight. (2) The subsystem must store the result of each measurement on board with its coordinates and a timestamp and keeps it until the base station confirms. (3) The subsystem must display all readings on a map within 5 min of packet arrival and flags test points outside user-set ranges.	(1) The robot will be placed 500 m away from the base station and 20 test measurement results will be sent to the base station it should successfully read 90% of the messages. (2) The robot will store 5 test packets, then be tested against what the base station stored. (3) The robot will send 5 test packets to the base station, and the base station's heatmap update response will be timed. Three of the test packets will have results outside of a pre-set range.
Soil Testing	(1) The subsystem must report pH, Electrical Conductivity, moisture, and temperature at each insertion. The output should be accurate to ±0.5 pH and ±1 °C. (1) The subsystem must complete a measurement in ≤ 10 min. (2) The subsystem must stop the probe within 10 mm of travel after an obstruction is detected, then retract.	(2) The robot will be instructed to lower the probe and read its output. (3) The robot will be instructed to test a point and be timed from the stopping of movement to the beginning of movement to the next point. (4) The probe will be inserted into a solid surface and the amount that continues to be extruded will be measured.
Drive Train / Chassis	(1) The subsystem must provide a device ≤ 30 kg. (2) The subsystem must climb a 15% grade in loose soil at full mass. (3) The subsystem must provide ≥ 8 cm ground clearance at full mass.	(1) Once the robot is fully set up, it will be weighed. (2) Dirt inclines at varying slopes will be created from 5%, 10%, 15%, to 20%. The robot will then have to climb at least the first three. (3) After being fully weighed down with all the components, the robot will be placed in loose dirt and will be measured from the bottom of the chassis to the dirt below.

Power / Charging	(1) The subsystem must collect ≥ 1.5 hr charge in 1 hr of sun. (2) The subsystem must hold ≥ 12 hr charge. (3) The subsystem must send each component its rated voltage ± 20%.	(1) The robot will be left in plain sun for 1 hr and the change of charge the of battery will be measured. (2) From a full charge, the robot will be turned on, and the time until the robot shuts off will be measured. (3) The battery will be charged to 25%, 50%, 75%, and 100%. The voltage supplied to each component will be measured and compared to its rating.

2.2. Constraints

The following constraints define the boundaries and limitations within which DIRTT must be designed and developed. Each constraint influences specific design decisions and ensures the final prototype remains functional, reliable, and feasible to produce within the project's scope. Table 2-3 lays out the type and description of each.

Table 0-3: Constraints

Type	Name	Description
Economic	Cost	The total budget of the project is $1000.
Operation	Size	The rover is to fit in an area of 24” wide, 24” deep, and 36” tall.
Environmental	Weather Protection	The enclosure must protect the electronics from rain, dust, and soil splash, meeting at least an IP65 rating [14].
Energy	Power Source	The system must run on battery power with solar charging, since grid power is not available in the field.
Operation	Communication	Data is limited to the range and data rate of the LoRa radio, which must comply with FCC Part 15 limits in the 915 MHz band [15].

Economic and time constraints have the most influence on the design process. With a total project budget of $1,000 and a two-semester timeline, the team must prioritize component selection based on performance-to-cost ratio. These constraints favor off-the-shelf sensors and modules over custom hardware and require realistic project pacing.

The environmental and energy constraints focus on keeping the system operating reliably outdoors. The enclosure must withstand dust, rain, and field debris, which directly affects material selection and sealing of the probe and cable openings. Because the system runs without grid power, the battery and solar panel must be sized to complete a full sampling run, which in turn limits the size and weight of the platform. The LoRa link provides long range but low data rates, so the system transmits compact sensor readings rather than large data files.

2.3. Standards

The design and development of D.I.R.T.T. requires adherence to recognized engineering and safety standards to ensure reliable, safe operation in an outdoor environment. These standards, shown in Table 2-4, guide the team in selecting weather-resistant enclosures, maintaining compliant wireless communication, and validating soil measurements. The standards below were selected based on their relevance to the system's electrical, mechanical, and communication components.

Table 0-4: Engineering Standards

Specific Standard	Standard Document	Specification / Application
IEC 60529	International Electrotechnical Commission (IEC) Standard: Degrees of Protection Provided by Enclosures (IP Code) [14]	Defines the IP rating system used to specify enclosure protection against dust and water, ensuring the electronics survive rain, irrigation, and field debris.
FCC 47 CFR Part 15, Subpart C (§15.247)	Federal Communications Commission (FCC) Regulation: *Operation Within the Bands 902–928 MHz, 2400–2483.5 MHz, and 5725–5850 MHz* [15]	Governs unlicensed spread-spectrum operation, including power output and channel use, for the system's LoRa radio transmission in the 915 MHz ISM band.
ASTM D2216	ASTM International Standard: *Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass* [16]	Provides the oven-dry reference method used to calibrate and verify the soil moisture sensor readings.

Collectively, these standards establish a foundation for durable, compliant, and accurate field-deployed design. Following these guidelines ensures that the enclosure, wireless link, and sensing hardware meet industry expectations for environmental protection, regulatory compliance, and measurement reliability. Compliance with these standards not only enhances user trust but also supports the long-term reliability of the final product.

2.4 REFERENCES

[1] United Nations, "Losing 25,000 to hunger every day," UN Chronicle, vol. 45, no. 2–3, Jun. 2008. [Online]. Available: https://www.un.org/en/chronicle/article/losing-25000-hunger-every-day

[2] USDA Economic Research Service, "U.S. agricultural import values outpaced export values in 2025,"USDA ERS, Jun. 30, 2026. [Online]. Available: https://www.ers.usda.gov/data-products/chart-gallery/58310

[3] A. B. Rosenberg and S. Wallander, "Farmer-reported soil and water concerns vary by crop type and region," USDA Economic Research Service, Amber Waves, May 6, 2022. [Online]. Available: https://www.ers.usda.gov/amber-waves/2022/may/farmers-report-soil-related-resource-concerns-on-bout-half-of-soybean-wheat-cotton-and-oat-fields

[4] NC State Extension, "Soil sampling strategies for site-specific field management," NC State Extension Publications. [Online]. Available: https://content.ces.ncsu.edu/soil-sampling-strategies-for-site-specific-field-management

[5] University of Minnesota Soil Testing Laboratory, "Lawn & garden soil testing services and fees," Univ. of Minnesota, St. Paul, MN, USA, 2026. [Online]. Available: https://soiltest.cfans.umn.edu/testing-services/lawn-garden. Accessed: Oct. 1, 2026.

[6] UW–Madison Division of Extension, Door County, "Lawn and garden soil sampling," Sturgeon Bay, WI, USA, 2026. [Online]. Available: https://door.extension.wisc.edu/files/2026/07/LAWN-AND-GARDEN-SOIL-SAMPLING-instructions-1.pdf. Accessed: Oct. 1, 2026.

[7] B. Barker, "Autonomous soil sampling: Robo sampling improves accuracy, "Top Crop Manager, https://www.topcropmanager.com/autonomous-soil-sampling-robo-sampling-improves-accuracy/

[8] G. Kitić et al., "Agrobot Lala—an autonomous robotic system for real-time, in-field soil sampling, and analysis of nitrates," Sensors, vol. 22, no. 11, p. 4207, May 2022. Doi:10.3390/s22114207

[9] E. Bönecke et al., "Guidelines for precise lime management based on high-resolution soil pH, texture and SOM maps generated from proximal soil sensing data," *Precis. Agric.*, vol. 22, pp. 493–523, 2021. [Online]. Available: https://oa.tib.eu/renate/handle/123456789/6365. Accessed: Oct. 1, 2026.

[10] Sugar Research Australia, "Soil apparent electrical conductivity," Sugar Research Australia, Brisbane, QLD, Australia, Information Sheet IS14016, 2025. [Online]. Available: https://sugarresearch.com.au/wp-content/uploads/2025/04/IS14016-Soil-Apparent-Electrical-Conductivity.pdf. Accessed: Oct. 1, 2026.

[11] D. Mori, et al., "Calibration complexities: Full-scale error imp[UR1]act and simultaneous variation of salinity, temperature, and moisture content on sensor performance in soil," Indian Inst. Technol. Mandi, Mandi, India. [Online]. Available: https://scene.iitmandi.ac.in/pages/publication/publication_details/760. Accessed: Oct. 1, 2026.

[12] "Variable rates and your soil," *Grainews*. [Online]. Available: https://www.grainews.ca/machinery/variable-rates-and-your-soil. Accessed: Oct. 1, 2026.

[13] New Mexico State University Cooperative Extension Service, "Soil biological processes," Las Cruces, NM, USA, Guide A153. [Online]. Available: https://pubs.nmsu.edu/_a/A153. Accessed: Oct. 1, 2026.

[14] International Electrotechnical Commission, "Degrees of protection provided by enclosures (IP Code)," IEC 60529:1989+AMD1:1999+AMD2:2013, Int. Electrotech. Comm., Geneva, Switzerland. [Online]. Available: https://webstore.iec.ch/en/publication/2452. Accessed: Oct. 6, 2026.

[15] Federal Communications Commission, "Operation within the bands 902-928 MHz, 2400-2483.5 MHz, and 5725-5850 MHz," 47 C.F.R. § 15.247, Fed. Commun. Comm., Washington, DC, USA. [Online]. Available: https://www.ecfr.gov/current/title-47/section-15.247. Accessed: Oct. 6, 2026.

[16] ASTM International, "Standard test methods for laboratory determination of water (moisture) content of soil and rock by mass," ASTM D2216-19, ASTM Int., West Conshohocken, PA, USA. [Online]. Available: https://www.astm.org/d2216-19.html. Accessed: Oct. 6, 2026.